COMPASS Synoptic CB: Sulfide

Sept 2026 Maryland Samples

2026-10-13

Code Set up

run information

```
##Before or After Reruns (YES or NO)
After_Reruns = "YES"
#If NO will not write final data file
  #And Will Write a Rerun File

###things that need to be changed
sample_month = "Sept"
sample_year = "2026"
Sample_Month_Year = "202609"
Run_Date = as.Date(c("10/01/2026"), format = "%m/%d/%Y")
Run_by = "Zoe Read"#Instrument user
Script_run_by ="Zoe Read"#Code user

Run_notes="Matrix checks and spikes look bad. Standards were 4 sd higher than usual, likely because we use

#Plate Data
folder_path_rawdata <- paste0("Raw Data")
Raw_plates <- list.files(path = folder_path_rawdata, full.names = TRUE,pattern = "Plate.*20261001\\.csv")
(Raw_plates)
```

```
## [1] "Raw Data/COMPASS_H2S_Sept2026_Plate1_20261001.csv"
## [2] "Raw Data/COMPASS_H2S_Sept2026_Plate2_20261001.csv"
## [3] "Raw Data/COMPASS_H2S_Sept2026_Plate3_20261001.csv"
## [4] "Raw Data/COMPASS_H2S_Sept2026_Plate4_20261001.csv"
```

```
# Define the file path for QAQC log file - NO Need to change just check year
log_path <- "Raw Data/Sulfide_STD_QAQC.csv"

# Metadata path
Raw_Metadata <- "Raw Data/COMPASS_SynopticCB_PW_SampleLog_2026.csv"

# Define final path for data be sure to change year and month and run number
final_path <- "Processed Data/COMPASS_SynopticCB_H2S_202609_MD_processed.csv"
rerun_path <- "Processed Data/COMPASS_SynopticCB_H2S_202609_MD_reruns.csv"
```

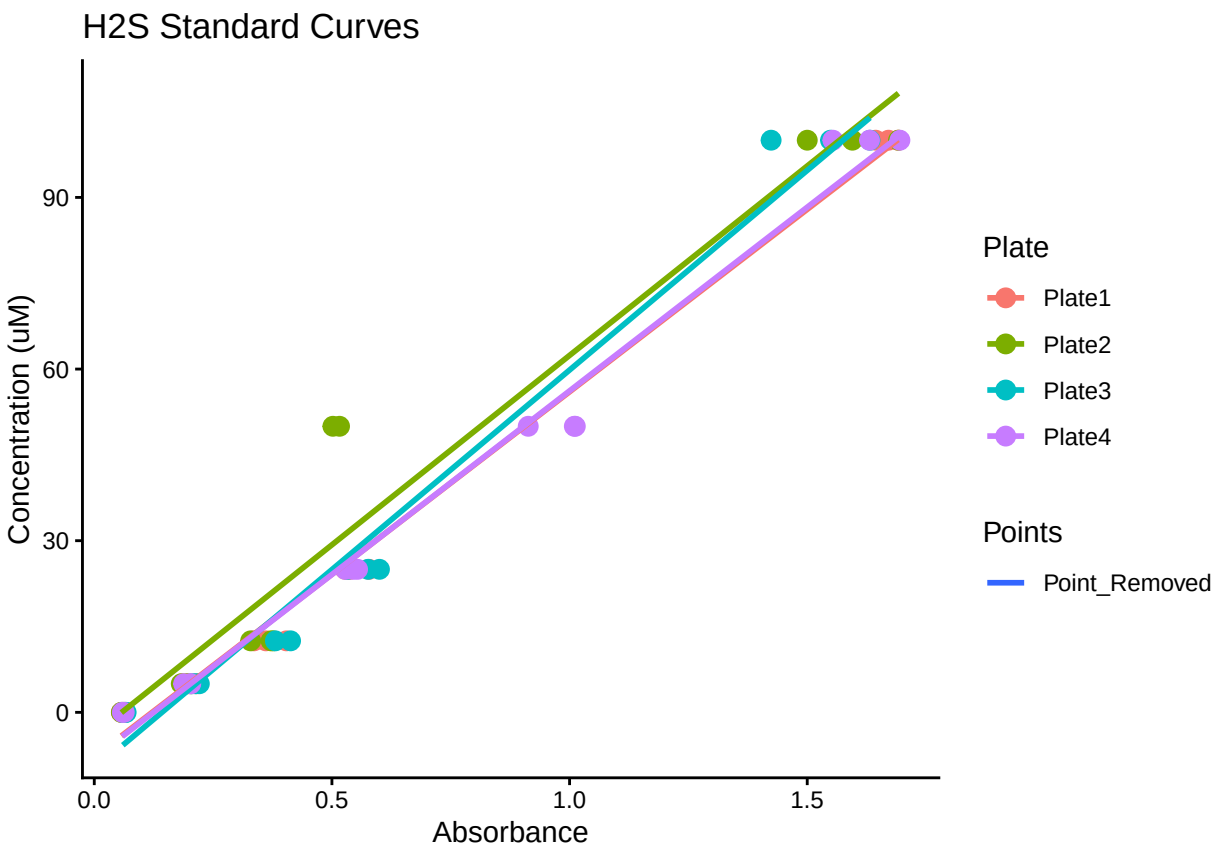
##Read in data

##Remove High CV Stds

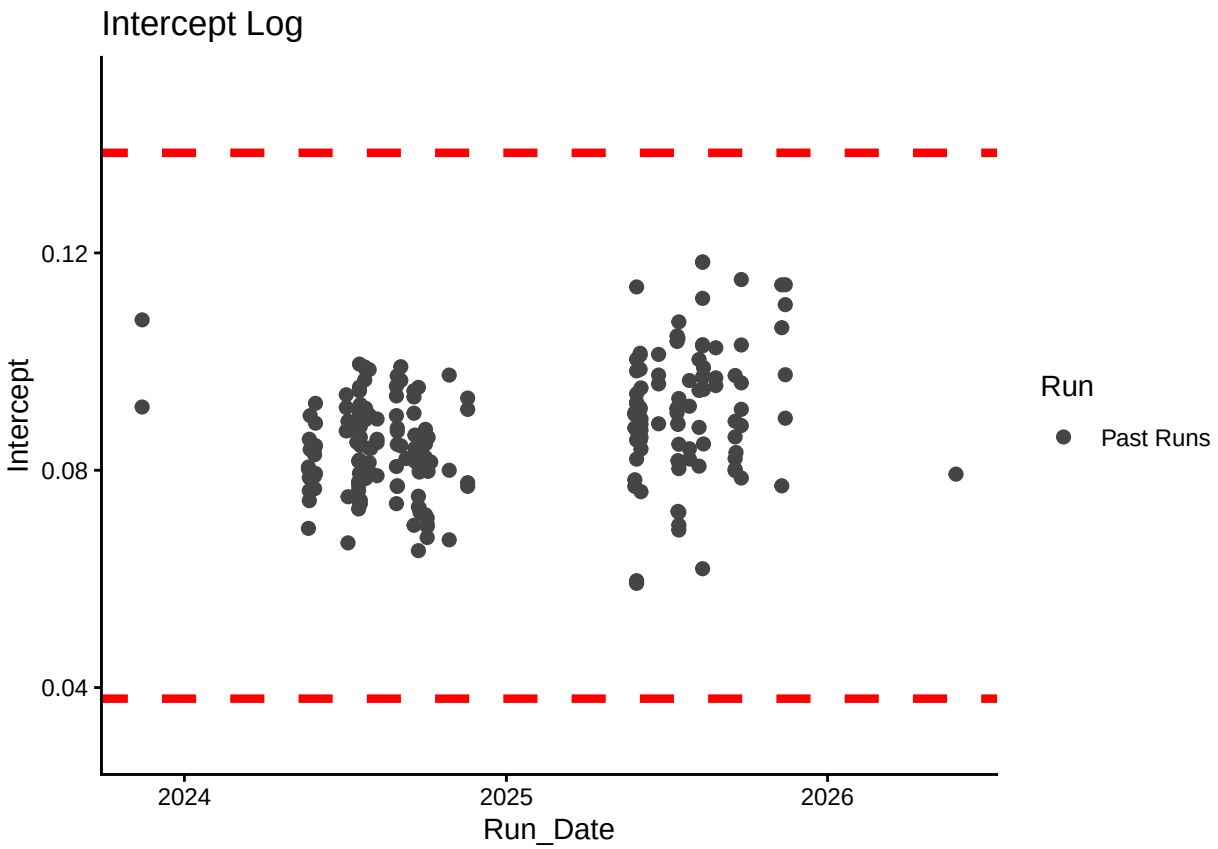
##Check R2 & Remove Standards if Necessary

```
## Joining with `by = join_by(Plate, IDs)`
```

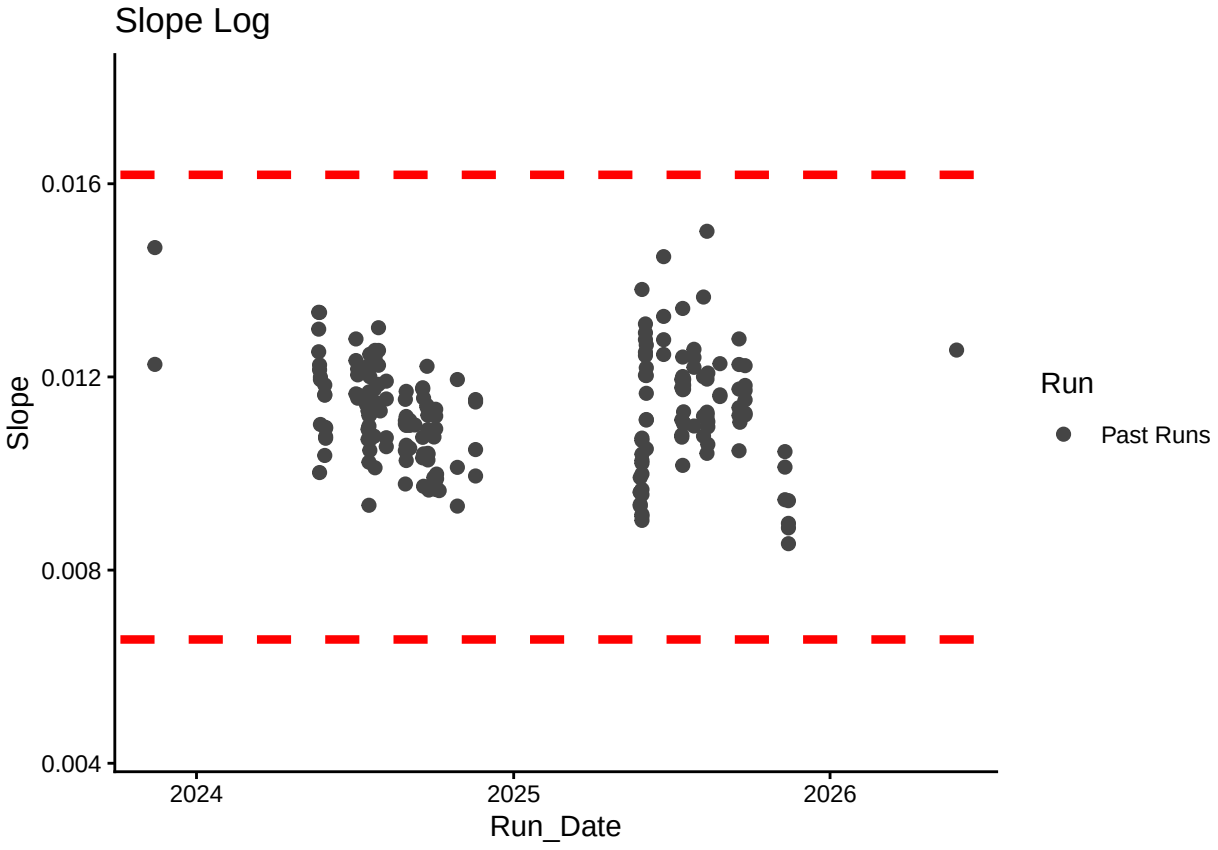
Plot Standard Curves




```
## Warning: Removed 17 rows containing missing values or values outside the scale range
## (‘geom_point()‘).
```




```
## Warning: Removed 17 rows containing missing values or values outside the scale range
## (`geom_point()`).
```

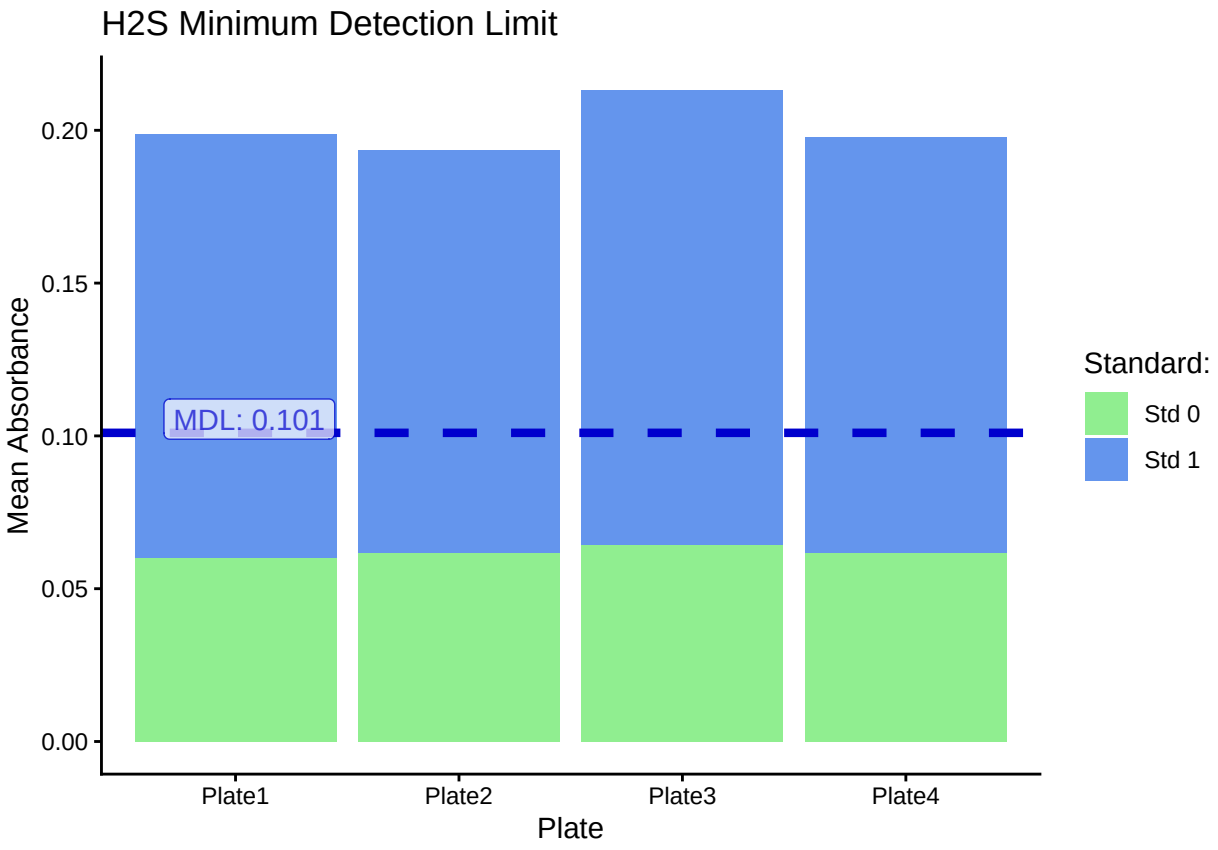


Return Flags For Standard Curve

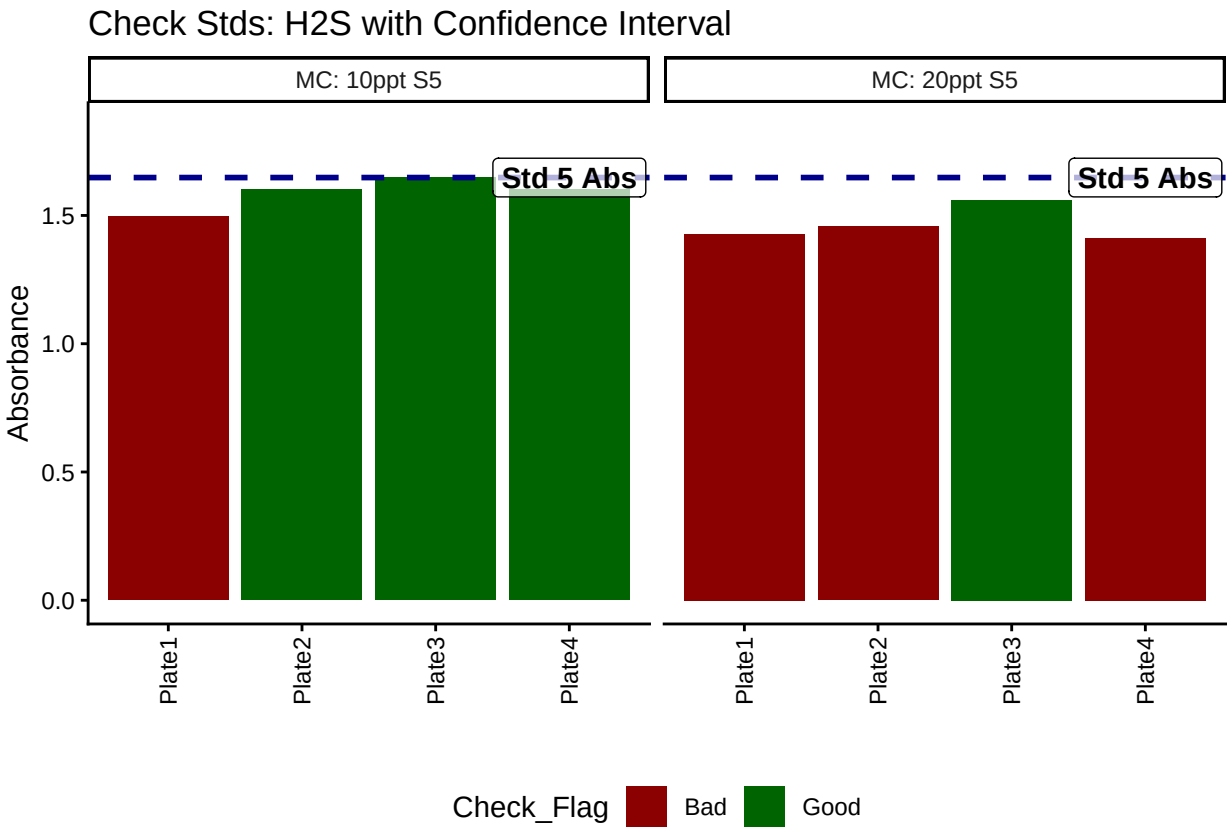
Plate	R2_Flag	CV_Flag
Plate1	R2_Good	CV_Good
Plate2	R2_Bad	CV_Good
Plate3	R2_Bad	CV_Good
Plate4	R2_Good	CV_Good

Plate	Curve_RMVD
Plate1	KEPT
Plate2	REMOVED
Plate3	REMOVED
Plate4	KEPT

Method Detection Limit



Compare Check Standards to Standards



Display Any Check Flags

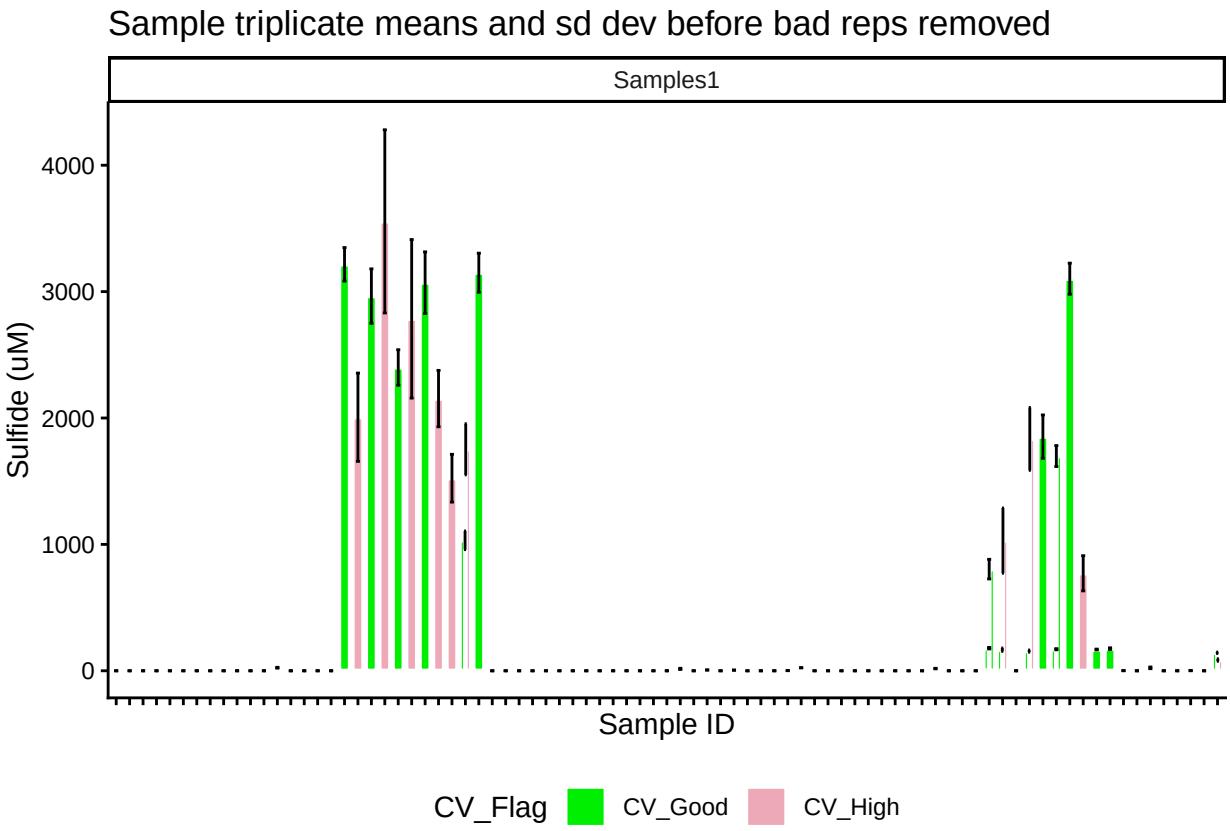
Plate	CHKs_Flag
Plate1	MC: 10ppt S5 Chk_Bad
Plate1	MC: 20ppt S5 Chk_Bad
Plate2	MC: 20ppt S5 Chk_Bad
Plate4	MC: 20ppt S5 Chk_Bad

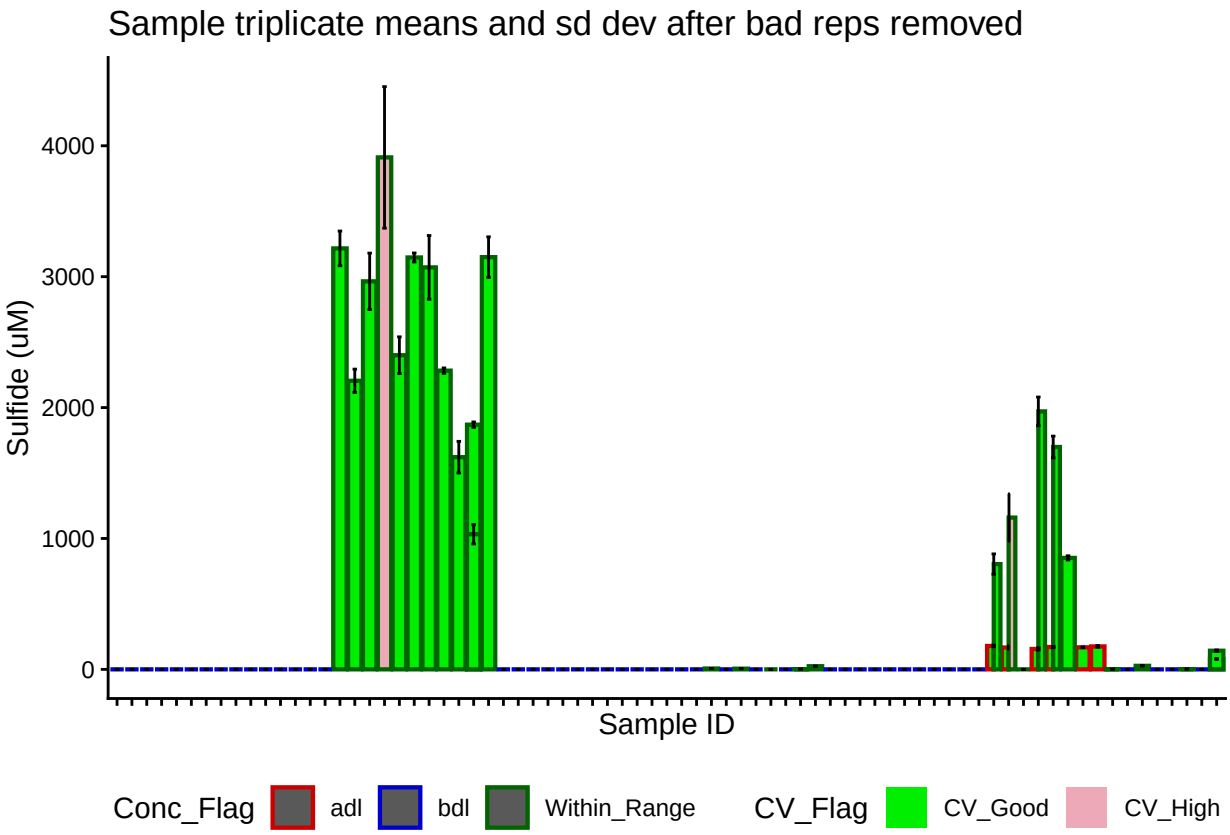
Plate	CHK_CV_Flags

##Calculate Sulfide Concentrations & Add Concentration Flags

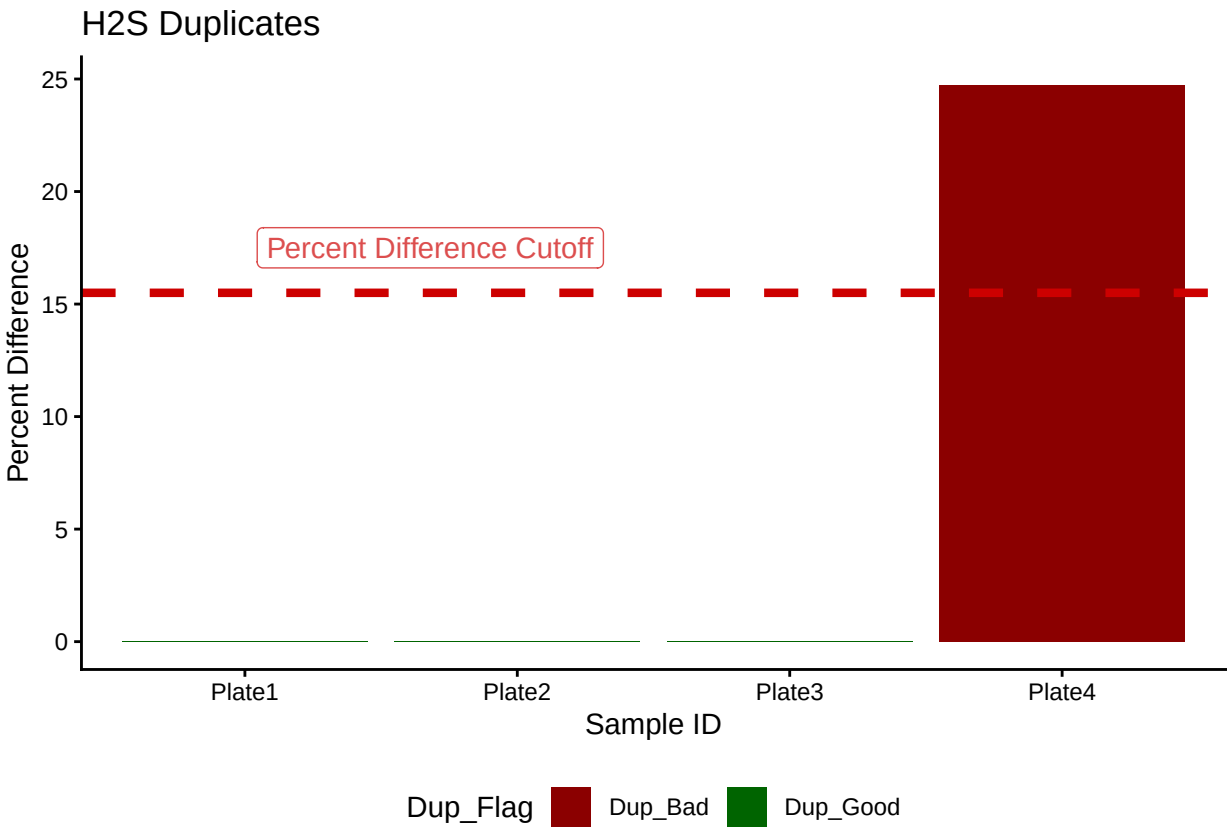
##Remove High CV Samples and Average

Plot Samples





Duplicate QAQC

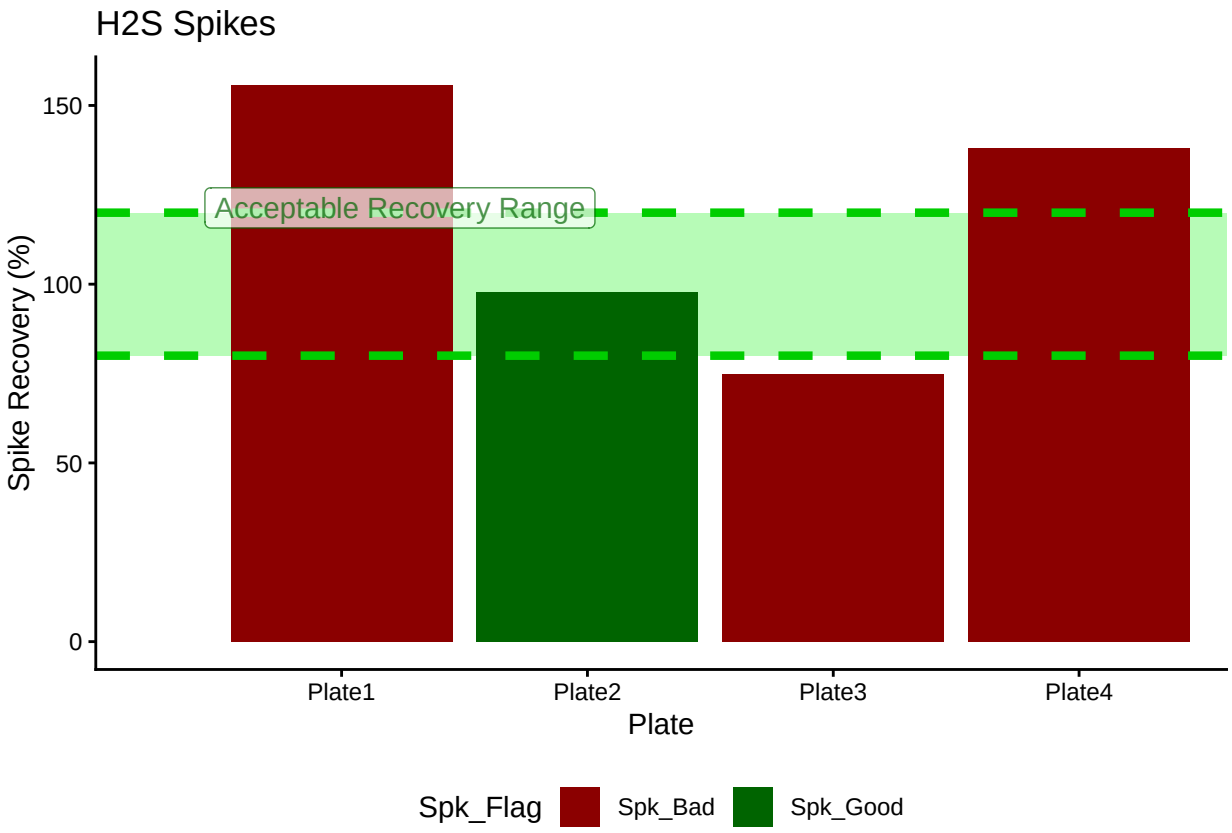


Display Dup Flags

```
## [1] ">60% Dups Pass"
```

Plate	Dup_Flag
Plate1	Dup_Good
Plate2	Dup_Good
Plate3	Dup_Good
Plate4	Dup_Bad

Spike QAQC



Display Spike Flags

```
## [1] "Spikes Paired"
```

Plate	Spike_Flag
Plate1	Spk_Bad
Plate2	Spk_Good
Plate3	Spk_Bad
Plate4	Spk_Bad

```
## [1] "<60% of Spikes Pass, Rerun"
```

Check for reruns and replace

##Add Flags

##Remove TEMPEST and FATE samples

```
Sample_data_IDs <- df_all_clean1 %>%
  filter(!str_detect(Sample_ID, "TMP")) %>%
  filter(!str_detect(Sample_ID, "FATE"))  %>%
  filter(!str_detect(Sample_ID, "dup"))  %>%
  filter(!str_detect(Sample_ID, "spk")) %>%
  filter(!str_detect(Sample_ID, "JULY"))
```

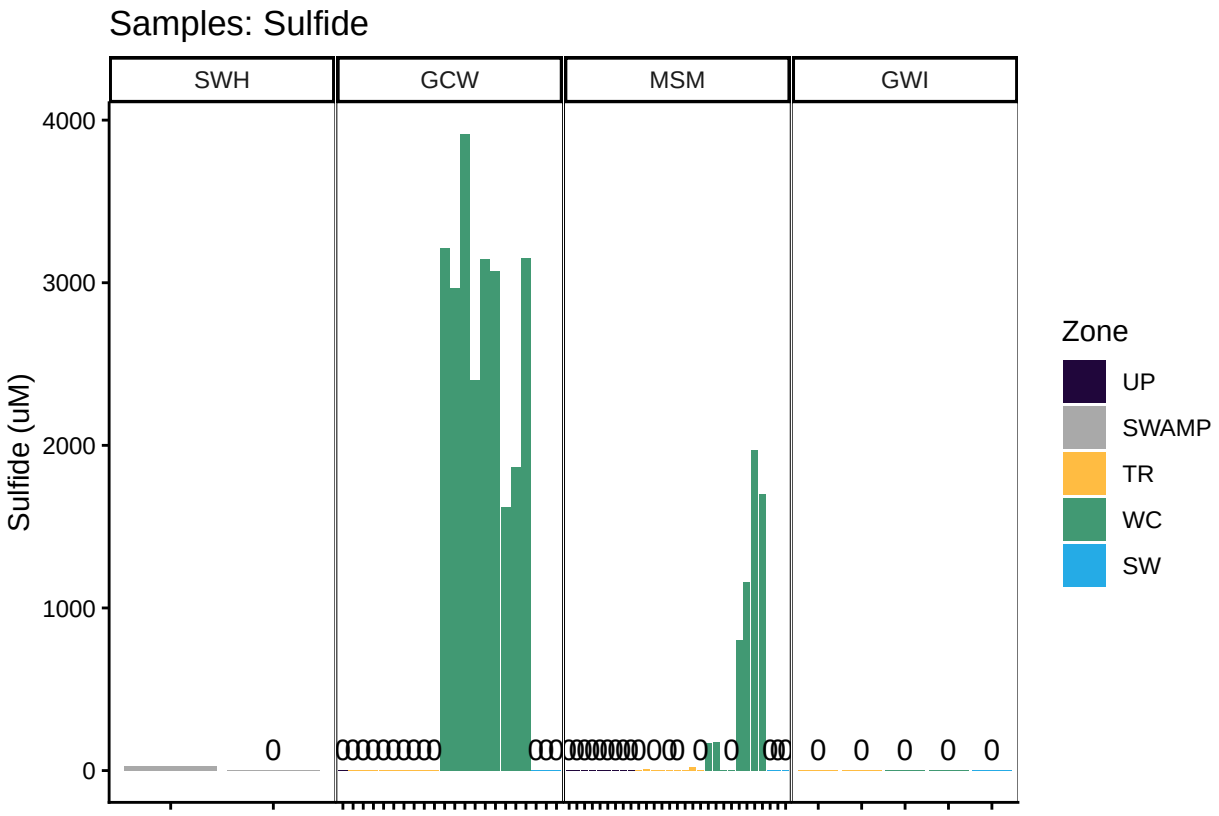
```r
July_samples <- df_all_clean1 %>%
  filter(str_detect(Sample_ID, "JULY")) %>%
  mutate(
    Project = "COMPASS: Synoptic",    # new column with same value on every row
    Region = "CB",
    Run_notes = Run_notes
  ) %>%
  rename(
    H2S_mean_uM = Conc,
    H2S_sd_uM = SD,
    H2S_cv = CV,
    H2S_conc_flag = Conc_Flag,
    H2S_cv_flag = CV_Flag,
    H2S_QAQC_flag = H2S_Flags,
    Analysis_rundate = Run_Date,
  ) %>%
  dplyr::select(
    Project, Region, Sample_ID,
         H2S_mean_uM, H2S_sd_uM, H2S_cv, H2S_conc_flag, H2S_cv_flag, H2S_QAQC_flag, Dilution,
         Analysis_rundate, Run_notes
        # list columns in the order you want them
  )

write.csv(July_samples, "../Sulfide Comparison/July_samples_timing_test.csv")
```

Check to see if samples run match metadata & merge info

```
## ***All sample IDs are present in metadata.***
```

Visualize Data by Plot



###END